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Cyber Crimes in Urban Bengaluru: Emerging Threats and Countermeasures

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Acknowledgement

Working on Cybercrime in Urban Bengaluru has been more than just an academic task — it's been a deep dive into a subject that's as real as it is relevant. From long nights of research to moments of reflection on the digital lives we lead, this paper has been shaped by many people along the way.

First, I want to thank Ms. Banu M, who didn't just guide this process but truly listened, questioned, and encouraged me through every twist and turn. Her patience and openness to my ideas made this work more thoughtful and grounded.

To St. Joseph's University, Bangalore, thank you for creating a space where tough questions can be asked, and new ideas can take shape — even the messy, uncomfortable ones that come with studying cybercrime in a fast-changing world.

I'm also grateful to the individuals and communities who are working every day to understand and fight cybercrime — from ethical hackers and digital safety advocates to police officials and researchers. Their stories and efforts added real-world weight to these pages.

And finally, to everyone navigating the complexities of life online — this paper is for you. Your experiences are what made this topic worth exploring.

Sagar G

Table of Contents

Abstract

1. Introduction

2. Literature Review

- 2.1. Cybercrime in India: An Analysis of Crime Against Women in Ever-Expanding Digital Space**
- 2.2. Cyber Crime and Cyber Laws in India**
- 2.3. Cyber Crime Threats and Security Legislations in India: A Critical Analysis**
- 2.4. Challenges in Cybercrime Prevention and Legal Frameworks in India**
- 2.5. Cybercrime in the Digital Age: Emerging Threats in India**
- 2.6. An Empirical Analysis of Cybercrime Trends in India (2017–2022)**
- 2.7. State-Level Policing Strategies for Cybercrime in India**
- 2.8. Cyberbullying in Indian Universities: A Study on Awareness and Prevention**
- 2.9. India's Cybersecurity Laws vs. International Standards: A Comparative Analysis**
- 2.10. Evolution of Cybercrime Laws in India: A Historical Perspective**

3. Objectives

4. Existing Cybercrime Landscape in Bengaluru

- 4.1. OLX Scam: Buy & Sell Fraud**
- 4.2. Trading Scam**
- 4.3. Phishing Attacks**
- 4.4. AI-Based Voice and Face Fraud**
- 4.5. WhatsApp Video Call Extortion**
- 4.6. Placement Scam**
- 4.7. Instagram Share Scam**
- 4.8. OTP Scam**
- 4.9. Delivery Scam**
- 4.10. Loan/Credit Card Scam**
- 4.11. Digital Arrest Scam**
- 4.12. UPI Fraud**

5. Methodology

- 5.1. Survey Results**

5.2. Statistical Analysis

6. Challenges and Limitations

- 6.1. Lack of Public Awareness Data Collection**
- 6.2. Evolving Cybercrime Tactics**
- 6.3. Delayed Law Enforcement Response**
- 6.4. Legal and Regulatory Gaps**
- 6.5. Financial and Emotional Impact on Victims**
- 6.6. Limited Cybersecurity Measures in Small Businesses**

7. Conclusion

8. References

Abstract

Cybercrime has seen an exponential rise in urban Bengaluru, affecting individuals, businesses, and banks and other financial services. The rapid adoption of digital payment systems, online marketplaces, and artificial intelligence (AI) tools has provided cybercriminals with new avenues to exploit vulnerabilities. This report looks into the evolving landscape of cybercrime in Bengaluru, focusing on recent trends such as OLX scams, AI-based voice and face fraud, WhatsApp video call extortion, placement scams, and digital arrests. These crimes not only result in financial losses but also can seriously affect personal privacy and security.

By looking at case studies, crime patterns, and law enforcement responses, this study highlights the tactics used by cybercriminals and the weaknesses in current digital security measures. It also explores the effectiveness of cybersecurity laws, efforts like Cybercrime.gov.in and the 1930 helpline, and awareness campaigns aimed at educating the public about cyber threats. It further shares recommendations for enhancing digital security, improving law enforcement responses, and fostering a cyber-aware society to combat the growing menace of cyber fraud in urban Bengaluru.^[1]

1. Introduction

Bengaluru, known as the Silicon Valley of India, is renowned for its thriving IT sector, innovation ecosystems, and rapid digital transformation. As one of India's foremost urban technology hubs, the city has witnessed unprecedented growth in digital infrastructure, online commerce, and financial technologies. However, this wave of digitalization has also made the city a prominent target for cybercriminal activities. With the proliferation of smartphones, widespread internet access, and the increasing reliance on digital services, residents of urban Bengaluru are more connected than ever before. Unfortunately, this connectivity has brought with it a surge in cyber threats that exploit technological vulnerabilities, human error, and gaps in public awareness.

In recent years, the nature of cybercrime in Bengaluru has evolved significantly. What once consisted of basic phishing emails and rudimentary hacking attempts has transformed into a sophisticated landscape of digital fraud. Criminals now employ advanced tactics such as AI-generated voice and face impersonations, deepfake technologies, and highly targeted social engineering attacks. Notable scams plaguing the city include OLX frauds involving fake buyers or sellers, WhatsApp video call extortion, trading scams promising unrealistic returns, and digital arrest schemes where scammers impersonate law enforcement officials to coerce payments from victims. Each of these schemes reflects a growing trend toward more complex, manipulative, and technologically savvy cybercrimes.

Among the most alarming developments is the use of artificial intelligence and deepfake technologies to facilitate crimes. AI-driven voice cloning and facial impersonation allow attackers to convincingly mimic trusted individuals, deceiving even the most vigilant targets. Such tools are commonly employed in business email compromise (BEC) schemes, fake recruitment campaigns,

and fraudulent investment opportunities propagated through social media platforms. As these technologies become more accessible, the challenge of distinguishing genuine interactions from malicious fabrications intensifies, particularly for individuals and small businesses lacking cybersecurity training.

The vulnerability of urban residents is further exacerbated by low levels of cybersecurity awareness. Despite numerous campaigns and the establishment of dedicated helplines like the 1930 Cyber Crime Helpline and online portals such as Cybercrime.gov.in, many individuals remain unaware of how to protect themselves online or respond to cyber threats. Studies show that even when victims recognize that they have been targeted, delays in reporting and a lack of technical knowledge often prevent effective action. This lack of awareness, coupled with the emotional and financial toll of cyber fraud, leaves citizens disempowered in the face of mounting digital risks.

Law enforcement agencies in Bengaluru are grappling with the rapid pace at which cyber threats are evolving. The technical nature of these crimes requires specialized skills, tools, and coordination between multiple jurisdictions. While Bengaluru has taken significant strides in establishing cybercrime cells and improving legal frameworks, enforcement efforts are often hampered by the transnational nature of cybercriminal networks, legal ambiguities in dealing with AI-driven offenses, and insufficient staffing and resources. Moreover, there is a pressing need to bridge the gap between the public and the police through proactive engagement, educational programs, and trust-building initiatives.

The current landscape of cybercrime also reveals that small and medium-sized enterprises (SMEs) are particularly vulnerable. Many lack dedicated IT departments or cybersecurity protocols, making them prime targets for data breaches, ransomware attacks, and phishing campaigns. Financial losses incurred through such attacks can be devastating, potentially shutting down operations or severely damaging reputations. For Bengaluru, a city driven by a vibrant entrepreneurial spirit and startup ecosystem, the resilience of these enterprises is critical not only for economic stability but also for fostering a culture of digital security.

This report seeks to address these urgent challenges by analyzing survey data from residents of urban Bengaluru. The survey focuses on gauging public awareness, understanding online behavior, identifying common threats, and evaluating the effectiveness of current cybersecurity practices. By combining statistical analysis with contextual insights from recent cybercrime trends, the report aims to provide a comprehensive overview of the digital safety landscape in the city.

It not only tries to map existing vulnerabilities but also to inform targeted interventions by stakeholders including government bodies, educational institutions, law enforcement agencies, and technology firms. As the digital footprint of urban Bengaluru continues to expand, it is imperative to foster a cyber-aware society that is resilient, informed, and prepared to navigate the complexities of the modern threat landscape. This introduction lays the foundation for the chapters that follow, which explore the specifics of public perception, cybersecurity habits, and policy recommendations to enhance digital safety in one of India's most connected cities.^[1]

2. Literature Review

2.1. “Cybercrime in India: An Analysis of Crime Against Women in Ever-Expanding Digital Space” – Sankhwar, S., et al. (2024)

This study uses a quantitative approach analyzing NCRB cybercrime data (2015–2023) with predictive modeling to understand the rising prevalence of cybercrimes against women. It focuses on threats such as cyberstalking, identity theft, and online harassment. The paper provides critical insights into gender-specific vulnerabilities in digital spaces, which is particularly relevant in urban centers like Bengaluru where social media usage is high. However, the study lacks qualitative victim narratives and does not consider inter-state variations, limiting its broader applicability.^[3]

2.2. “Cyber Crime and Cyber Laws in India” – Thangamayan, S., et al. (2023)

This paper offers a doctrinal legal analysis of Indian cyber laws, international conventions, and judicial interpretations. It reviews the IT Act 2000 and its subsequent amendments, uncovering significant legal loopholes and challenges in enforcement. The study is valuable for understanding the structural and legal foundations of India’s cybersecurity ecosystem. In the context of cities like Bengaluru, where digital crimes are becoming increasingly complex, the paper underscores the urgency of updating legal frameworks. However, it is limited by its narrow focus on legal aspects, omitting practical technological and societal dimensions.^[4]

2.3. “Cyber Crime Threats and Security Legislations in India: A Critical Analysis” – Choudhary, V. N., & Srivastava, S. (2024)

This survey-based study collects data from cybercrime victims, law enforcement officers, and IT professionals to evaluate the effectiveness of India’s cybersecurity policies. It identifies gaps in public awareness and enforcement, making it particularly relevant for high-density urban regions like Bengaluru. The study highlights the need for better public education and stronger institutional collaboration. However, it focuses primarily on urban settings and lacks rural perspectives, which might affect the comprehensiveness of its policy recommendations.^[5]

2.4. “Challenges in Cybercrime Prevention and Legal Frameworks in India” – Nakkeeran, S., & Singh, D. (2024)

Utilizing a case study approach, this research analyzes 20 high-profile cybercrime incidents to understand how these crimes are prosecuted under Indian law. It provides valuable insights into the operational challenges faced by law enforcement agencies, especially in tech-heavy cities like Bengaluru. The paper emphasizes the gap between legal provisions and real-world enforcement, highlighting issues like jurisdictional overlap and technical resource shortages. However, its focus on high-profile cases limits its relevance to everyday cybercrimes.^[6]

2.5. “Cybercrime in the Digital Age: Emerging Threats in India” – Kasturi, Y., & Dar, M. A. (2024)

This mixed-methods study incorporates interviews with cybersecurity experts and content analysis of threat reports to identify emerging digital risks, such as AI-driven scams and deepfake technology. It is highly relevant to Bengaluru's rapidly evolving cybercrime landscape, where such sophisticated tactics are increasingly common. The paper offers foresight into the future of cyber threats, aiding policymakers and cybersecurity professionals in preparation. However, it relies heavily on expert opinions, which introduces a degree of subjectivity to its findings. ^[7]

2.6. “An Empirical Analysis of Cybercrime Trends in India (2017–2022)” – Shukla, V. (2023)

Based on secondary analysis of cybercrime reports and CERT-In data, this paper provides an empirical overview of key trends, including financial fraud, hacking, and cyberbullying. It is instrumental in establishing a statistical foundation for understanding the rise of cybercrime in India. The study's broad scope makes it a useful benchmark for city-specific investigations, including those focused on Bengaluru. However, it lacks forward-looking predictive models that could inform future risk mitigation strategies. ^[8]

2.7. “State-Level Policing Strategies for Cybercrime in India” – Attri, V. K., et al. (2025)

Drawing from field interviews with police officers and cybersecurity professionals across six Indian states, this study examines regional disparities in cybercrime enforcement strategies. It sheds light on what policing models work best and how they could be adapted for Bengaluru's cybercrime cells. The research is especially useful for comparing enforcement effectiveness across jurisdictions. Nonetheless, its findings are region-specific and may not be entirely generalizable to the national or pan-urban level. ^[9]

2.8. “Cyberbullying in Indian Universities: A Study on Awareness and Prevention” – Kaur, M., & Saini, M. (2023)

This qualitative research utilizes thematic analysis of interviews with students and faculty to investigate cyberbullying in higher education institutions. Given Bengaluru's large student population and vibrant academic culture, the study's insights into awareness levels and preventive policies are particularly relevant. It also highlights institutional gaps in addressing digital harassment. However, the exclusion of school students and working professionals limits the scope of its findings. ^[10]

2.9. “India's Cybersecurity Laws vs. International Standards: A Comparative Analysis” – Misra, A., & Chacko, M. (2021)

Through a comparative analysis of India's cybersecurity legislation and international standards like the Budapest Convention, this paper assesses the adequacy of India's legal framework in

addressing modern cyber threats. It provides a global context to India's cybersecurity evolution, which is essential for tech-forward cities like Bengaluru aiming to align with international best practices. Yet, the study does not explore practical enforcement challenges within the Indian context. ^[11]

2.10. “Evolution of Cybercrime Laws in India: A Historical Perspective” – Kethineni, S. (2019)

This historical analysis tracks the development of cybercrime legislation and policy responses in India from 2005 to 2019. It highlights the gradual institutional recognition of cyber threats and the legislative journey toward digital security. This background is crucial for understanding how India's current legal architecture came into being. However, the data and context are somewhat outdated, and the study does not address newer threats like AI-based fraud or deepfake scams increasingly relevant to Bengaluru. ^[12]

3. Objectives

- To analyze the various types of cyber frauds occurring in Bangalore Urban, including financial scams, social media fraud, and phishing attacks.
- To examine the role of emerging technologies like AI, deepfake, and blockchain in cyber frauds and their prevention.
- To evaluate the effectiveness of existing cybersecurity laws and regulations in combating digital crimes.
- To evaluate fraud patterns in urban and rural areas and compare the impact on different demographics.

4. Existing Cybercrime Landscape in Bengaluru

4.1. OLX Scam: Buy & Sell Fraud

Here, scammers fraudsters posing as buyers or sellers on platforms like OLX and other e-commerce websites. They use fake payment links or deceptive QR codes to trick victims into transferring money under the guise of completing a transaction. Often, the scammers impersonate trusted figures such as army personnel or government officials to build credibility and lower the victim's guard.

4.2. Trading Scam

Fraudulent online trading platforms lure victims with promises of high returns on investments in stocks, forex, or cryptocurrencies. These scams often rely on fake advertisements and operate similarly to Ponzi schemes, manipulating trust and urgency. After collecting substantial deposits from unsuspecting investors, the scammers abruptly disappear, leaving victims without recourse.

4.3. Phishing Attacks

Cybercriminals conduct phishing by sending emails, SMS, or WhatsApp messages that appear to come from legitimate institutions like banks or service providers. These messages contain links that redirect users to counterfeit websites designed to steal credentials and personal information. The aftermath often includes unauthorized financial transactions and identity theft.

4.4. AI-Based Voice and Face Fraud

Utilizing deepfake technologies, fraudsters can now impersonate individuals' voices and faces with startling accuracy. These AI-generated manipulations are used to deceive victims during phone calls or video interactions, often within business email compromise (BEC) scams. The impersonation of trusted individuals enables unauthorized access to sensitive data and funds.

4.5. WhatsApp Video Call Extortion

In this scheme, scammers initiate WhatsApp video calls and secretly record compromising footage of their victims. The victims are then threatened with fabricated legal consequences or public exposure unless they pay a ransom. This tactic typically targets working professionals and younger individuals active on social media platforms.

4.6. Placement Scam

Scammers create fake job portals and pose as recruitment agents to extract money from job seekers under the pretense of registration or training fees. Victims are sometimes issued fake offer letters claiming to be from reputable firms. Once the payment is made, the fraudsters vanish, and the job opportunity is revealed to be nonexistent.

4.7. Instagram Share Scam

Hackers take control of popular Instagram accounts and use them to promote fake investment schemes. Followers, believing the message is genuine, are encouraged to invest in these

fraudulent ventures. After funds are transferred, the scammers delete or abandon the hacked account, leaving investors defrauded.

4.8. OTP Scam

In this scam, fraudsters impersonate banks, courier companies, or online platforms and coax victims into revealing one-time passwords (OTPs). Once shared, these OTPs are exploited to gain unauthorized access to bank accounts, digital wallets, or personal social media profiles.

4.9. Delivery Scam

Fraudsters pretend to be delivery personnel and request OTPs from victims to confirm bogus deliveries. These OTPs are then misused to access and drain funds from payment wallets or banking apps. This scam is prevalent with services related to food delivery, e-commerce, and courier shipments.

4.10. Loan/Credit Card Scam

Scammers lure victims with offers of pre-approved loans or credit cards that require minimal verification. Victims are asked to pay processing or documentation fees upfront, after which the scammers disappear. In more severe cases, the personal data provided is exploited for further fraudulent activities.

4.11. Digital Arrest Scam

Fraudsters pretend to be law enforcement officers and claim that the victim is involved in illegal activity. They use fear tactics, such as threatening arrests or legal action, to pressure victims into paying fines or bribes. Spoofed IDs and fabricated legal documents are often used to lend authenticity to the scam.

4.12. UPI Fraud

In this fraud, sending fake UPI links via email, SMS, or messaging apps like WhatsApp. When clicked, the link executes unauthorized transactions from the victim's account. Scammers also trick victims into scanning malicious QR codes that withdraw funds instead of crediting them, taking advantage of UPI's convenience and speed.

5. Methodology

This study uses a mix of research methods to analyze the prevalence and impact of cybercrime in urban Bengaluru, combining both primary and secondary research methods. The main method used for this research is a structured survey conducted among residents of Bengaluru. The survey was designed to understand public awareness of cyber threats, frequency of exposure to cyber fraud, and the general understanding of cybersecurity practices. Participants were selected from diverse age groups, occupations, and regions within the city to ensure broad representation. The survey data provides a ground-level view of digital behavior, threat perception, and the effectiveness of personal cybersecurity measures.

In addition to the primary survey, secondary research methods were employed to enrich the analysis. Case study analysis was conducted by examining real-world cyber fraud incidents reported in Bengaluru. These cases helped in identifying recurring patterns and the common tactics used by cybercriminals in the region. This qualitative component offered context to the statistical insights from the survey.

Expert interviews were another critical component of the methodology. Cybersecurity professionals and law enforcement officers were consulted to gain deeper insights into the operational challenges in detecting, investigating, and prosecuting cybercrimes. These discussions shed light on institutional preparedness, response strategies, and gaps in current frameworks.

Government reports and official data were also reviewed, including records from Cybercrime.gov.in and the 1930 cyber helpline. This analysis helped evaluate the effectiveness of existing legal and support mechanisms available to the public. Together, these components contribute to a comprehensive understanding of Bengaluru's cybercrime landscape, enabling more targeted and effective recommendations.

5.1. Survey Results

What is your age group ?	What is your gender?	What is your occupation?	Which part of Bangalore do you live in?	How would you rate your awareness of cybersecurity ?	Are you familiar with terms like 'Phishing', 'Malware', etc.?	Have you received formal cybersecurity training?	How many hours do you spend online daily?	What do you use the internet for most?	Which cybersecurity practice do you follow most often?	Have you ever been a victim of cybercrime?	If yes, which of the following best describes it?	Did you report the cyber incident?	Do you think cybercrimes are increasing in urban Bangalore?	Should the government raise cybersecurity awareness ?	Should schools/colleges include cybersecurity education ?
18–25	Prefer not to say	Homemaker	East Bangalore	No Awareness	No, I don't know them	No	Less than 1 hour	Studying	Using antivirus	No	Identity theft	Yes	Strongly Disagree	Yes	Maybe
18–25	Female	Business Owner	Outside Bangalore	Low	Yes, I know them well	No	More than 6 hours	Work/ Business	Avoiding unknown links	No	Cyberbullying	Not applicable	Disagree	Not Sure	Maybe

36–50	Prefer not to say	Retired	West Bangalore	Low	No, I don't know them	Yes	1–3 hours	Online Shopping	Using antivirus	No	Hacking	Not applicable	Neutral	Not Sure	Maybe
18–25	Other	Business Owner	Central Bangalore	Very High	Yes, I know them well	Yes	More than 6 hours	Work/Business	Two-factor authentication	Not sure	Hacking	Not applicable	Strongly Agree	Yes	Maybe
Under 18	Male	Other	Outside Bangalore	Low	No, I don't know them	Yes	More than 6 hours	Studying	Strong passwords	No	Hacking	Not applicable	Neutral	Not Sure	No
18–25	Male	Student	North Bangalore	High	I've heard of them, but not sure	No	3–6 hours	Work/Business	Using antivirus	Not sure	None / Prefer not to say	No	Agree	Not Sure	Not Sure
Under 18	Other	Homemaker	Central Bangalore	High	Yes, I know them well	Not sure	3–6 hours	Work/Business	Two-factor authentication	No	Online scam/fraud	Not applicable	Disagree	Yes	No
18–25	Female	Student	Central Bangalore	Low	Yes, I know them well	No	Less than 1 hour	Work/Business	Using antivirus	Yes	None / Prefer not to say	Yes	Strongly Disagree	Not Sure	Maybe
18–25	Female	Retired	Outside Bangalore	Very High	No, I don't know them	Yes	3–6 hours	Work/Business	Using antivirus	Yes	Online scam/fraud	I didn't know where to report	Neutral	Not Sure	Not Sure
Under 18	Other	Student	East Bangalore	Low	No, I don't know them	No	1–3 hours	Banking /UPI	Avoiding unknown links	Not sure	Cyberbullying	No	Strongly Agree	Not Sure	Maybe
Under 18	Female	Business Owner	Central Bangalore	Very High	Yes, I know them well	Not sure	1–3 hours	Banking /UPI	Strong passwords	No	None / Prefer not to say	I didn't know where to report	Disagree	Not Sure	Maybe
18–25	Female	Business Owner	South Bangalore	Very High	No, I don't know them	Not sure	More than 6 hours	Online Shopping	Two-factor authentication	Not sure	Hacking	Yes	Neutral	Not Sure	Not Sure
18–25	Prefer not to say	Retired	Outside Bangalore	High	I've heard of them, but not sure	Not sure	More than 6 hours	Online Shopping	Avoiding unknown links	No	Online scam/fraud	No	Strongly Disagree	Not Sure	Yes
Under 18	Prefer not to say	Other	South Bangalore	No Awareness	I've heard of them, but not sure	No	1–3 hours	Work/Business	Using antivirus	Not sure	Hacking	Yes	Strongly Disagree	Yes	Yes
18–25	Other	Working Professional	East Bangalore	Very High	Yes, I know them well	Not sure	Less than 1 hour	Banking /UPI	Avoiding unknown links	No	Hacking	No	Strongly Disagree	No	No
36–50	Male	Student	East Bangalore	Moderate	Yes, I know them well	Yes	More than 6 hours	Banking /UPI	Strong passwords	Yes	Identity theft	No	Strongly Disagree	No	Yes
26–35	Prefer not to say	Retired	Central Bangalore	High	No, I don't know them	Yes	1–3 hours	Banking /UPI	Strong passwords	Yes	OTP/UPI fraud	Yes	Strongly Agree	No	Maybe
18–25	Other	Homemaker	Outside Bangalore	No Awareness	No, I don't know them	Not sure	3–6 hours	Social Media	None of the above	Yes	Online scam/fraud	Not applicable	Disagree	Yes	No
18–25	Prefer not to say	Homemaker	East Bangalore	Very High	I've heard of them, but not sure	Not sure	More than 6 hours	Banking /UPI	Using antivirus	Not sure	Hacking	Yes	Disagree	Not Sure	Maybe

36–50	Other	Business Owner	Outside Bangalore	Low	I've heard of them, but not sure	Not sure	Less than 1 hour	Entertainment	Two-factor authentication	No	Hacking	Not applicable	Strongly Agree	Yes	Maybe
Under 18	Prefer not to say	Business Owner	South Bangalore	Moderate	No, I don't know them	Not sure	3–6 hours	Entertainment	None of the above	Yes	Identity theft	Not applicable	Neutral	No	Yes
Under 18	Prefer not to say	Student	West Bangalore	Very High	Yes, I know them well	Yes	More than 6 hours	Social Media	None of the above	No	Identity theft	Not applicable	Agree	No	Not Sure
36–50	Female	Student	West Bangalore	Moderate	No, I don't know them	Not sure	3–6 hours	Online Shopping	Two-factor authentication	Yes	Hacking	Yes	Strongly Agree	Not Sure	No
Under 18	Prefer not to say	Homemaker	Outside Bangalore	Moderate	No, I don't know them	No	1–3 hours	Social Media	Avoiding unknown links	Not sure	None / Prefer not to say	Yes	Agree	Yes	Yes
51+	Female	Retired	North Bangalore	Low	No, I don't know them	Yes	1–3 hours	Work/Business	Two-factor authentication	No	Identity theft	I didn't know where to report	Strongly Disagree	Yes	Maybe
51+	Male	Homemaker	North Bangalore	Low	I've heard of them, but not sure	No	Less than 1 hour	Entertainment	Avoiding unknown links	Not sure	Cyberbullying	Not applicable	Strongly Disagree	Yes	Yes
18–25	Female	Retired	Central Bangalore	Very High	I've heard of them, but not sure	No	More than 6 hours	Work/Business	Two-factor authentication	Yes	Cyberbullying	I didn't know where to report	Strongly Disagree	No	Maybe
Under 18	Prefer not to say	Student	East Bangalore	No Awareness	Yes, I know them well	Yes	More than 6 hours	Studying	Two-factor authentication	Not sure	None / Prefer not to say	No	Agree	Yes	Yes
26–35	Male	Student	South Bangalore	Very High	Yes, I know them well	Yes	Less than 1 hour	Social Media	None of the above	Yes	None / Prefer not to say	I didn't know where to report	Agree	No	Maybe
26–35	Male	Other	West Bangalore	No Awareness	Yes, I know them well	Not sure	3–6 hours	Social Media	Strong passwords	Not sure	Cyberbullying	I didn't know where to report	Neutral	No	Yes
Under 18	Prefer not to say	Other	Central Bangalore	Very High	Yes, I know them well	No	More than 6 hours	Online Shopping	Using antivirus	Yes	None / Prefer not to say	Yes	Strongly Disagree	Not Sure	Yes
26–35	Other	Retired	Outside Bangalore	Moderate	Yes, I know them well	No	1–3 hours	Social Media	None of the above	No	Online scam/fraud	I didn't know where to report	Agree	Yes	No
36–50	Female	Retired	East Bangalore	Moderate	Yes, I know them well	Not sure	3–6 hours	Banking /UPI	Using antivirus	Not sure	Identity theft	No	Disagree	No	Not Sure
51+	Other	Retired	North Bangalore	Very High	Yes, I know them well	No	More than 6 hours	Online Shopping	Using antivirus	No	Hacking	No	Agree	No	Maybe
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Under 18	Prefer not to say	Student	West Bangalore	High	No, I don't know them	Yes	More than 6 hours	Studying	Avoiding unknown links	Yes	Identity theft	Not applicable	Strongly Disagree	Not Sure	No
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26–35	Other	Working Professional	Outside Bangalore	Moderate	No, I don't know them	No	More than 6 hours	Banking /UPI	Two-factor authentication	Not sure	None / Prefer not to say	Not applicable	Agree	No	Maybe
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36–50	Male	Retired	North Bangalore	High	Yes, I know them well	Yes	3–6 hours	Online Shopping	None of the above	Not sure	Hacking	Yes	Disagree	Not Sure	Yes

51+	Other	Retired	Central Bangalore	No Awareness	Yes, I know them well	No	Less than 1 hour	Banking /UPI	Avoiding unknown links	No	Online scam/fraud	No	Strongly Disagree	Not Sure	Maybe
51+	Male	Homemaker	South Bangalore	No Awareness	I've heard of them, but not sure	No	1-3 hours	Entertainment	Two-factor authentication	Not sure	OTP/UPI fraud	I didn't know where to report	Neutral	No	Maybe
51+	Other	Homemaker	Outside Bangalore	Moderate	No, I don't know them	Yes	3-6 hours	Studying	Avoiding unknown links	Not sure	Hacking	Not applicable	Agree	No	Maybe
36-50	Prefer not to say	Retired	Outside Bangalore	Moderate	No, I don't know them	No	1-3 hours	Online Shopping	Avoiding unknown links	Yes	Cyberbullying	I didn't know where to report	Agree	Not Sure	No
18-25	Female	Business Owner	Central Bangalore	Very High	I've heard of them, but not sure	Yes	3-6 hours	Online Shopping	Avoiding unknown links	Not sure	Online scam/fraud	Yes	Strongly Agree	Yes	No
51+	Female	Working Professional	East Bangalore	Low	I've heard of them, but not sure	Not sure	1-3 hours	Social Media	Two-factor authentication	Not sure	Online scam/fraud	No	Strongly Agree	No	Maybe
26-35	Other	Business Owner	Outside Bangalore	Low	Yes, I know them well	Yes	More than 6 hours	Studying	Strong passwords	No	Hacking	No	Strongly Disagree	Not Sure	Not Sure
Under 18	Prefer not to say	Retired	North Bangalore	High	Yes, I know them well	Not sure	More than 6 hours	Studying	None of the above	No	Online scam/fraud	Not applicable	Strongly Disagree	Yes	Not Sure
36-50	Prefer not to say	Other	East Bangalore	Very High	I've heard of them, but not sure	Yes	3-6 hours	Banking /UPI	Avoiding unknown links	Yes	Hacking	Not applicable	Neutral	Not Sure	Maybe
51+	Other	Other	East Bangalore	High	I've heard of them, but not sure	Not sure	Less than 1 hour	Work/Business	Strong passwords	Yes	OTP/UPI fraud	No	Neutral	No	Yes
36-50	Female	Working Professional	Outside Bangalore	High	Yes, I know them well	Yes	Less than 1 hour	Work/Business	Strong passwords	No	Cyberbullying	Not applicable	Agree	Yes	Not Sure
51+	Other	Student	Outside Bangalore	Low	I've heard of them, but not sure	Not sure	1-3 hours	Work/Business	Strong passwords	Not sure	OTP/UPI fraud	Yes	Strongly Agree	Not Sure	No
36-50	Male	Business Owner	East Bangalore	No Awareness	I've heard of them, but not sure	No	3-6 hours	Social Media	Two-factor authentication	Yes	Hacking	I didn't know where to report	Strongly Agree	No	Not Sure
26-35	Female	Business Owner	East Bangalore	Moderate	No, I don't know them	No	More than 6 hours	Online Shopping	None of the above	No	None / Prefer not to say	Yes	Disagree	Not Sure	Not Sure

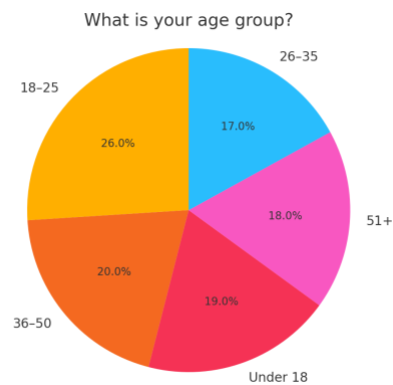
18–25	Female	Business Owner	East Bangalore	Low	I've heard of them, but not sure	No	More than 6 hours	Online Shopping	Avoiding unknown links	No	Identity theft	I didn't know where to report	Strongly Disagree	No	Yes
18–25	Other	Retired	East Bangalore	Very High	Yes, I know them well	No	1–3 hours	Work/Business	Strong passwords	No	OTP/UPI fraud	Yes	Strongly Disagree	No	Maybe
26–35	Other	Retired	Central Bangalore	Low	I've heard of them, but not sure	Yes	3–6 hours	Studying	None of the above	Not sure	Identity theft	Not applicable	Agree	Not Sure	Yes
36–50	Female	Retired	North Bangalore	No Awareness	No, I don't know them	Yes	1–3 hours	Online Shopping	None of the above	Yes	None / Prefer not to say	Yes	Agree	No	Maybe
26–35	Male	Homemaker	North Bangalore	High	I've heard of them, but not sure	No	More than 6 hours	Banking /UPI	Two-factor authentication	No	Online scam/fraud	I didn't know where to report	Strongly Agree	No	Yes
18–25	Other	Retired	East Bangalore	No Awareness	No, I don't know them	No	1–3 hours	Work/Business	Avoiding unknown links	Yes	OTP/UPI fraud	No	Strongly Agree	No	Yes
36–50	Prefer not to say	Working Professional	West Bangalore	Moderate	No, I don't know them	Yes	More than 6 hours	Banking /UPI	None of the above	Not sure	Cyberbullying	I didn't know where to report	Disagree	Not Sure	Maybe
51+	Other	Other	Outside Bangalore	Very High	Yes, I know them well	Not sure	More than 6 hours	Social Media	Avoiding unknown links	Yes	Cyberbullying	I didn't know where to report	Strongly Agree	Yes	Not Sure
Under 18	Male	Retired	Central Bangalore	Moderate	Yes, I know them well	No	1–3 hours	Online Shopping	Two-factor authentication	No	None / Prefer not to say	I didn't know where to report	Strongly Disagree	Yes	Yes
18–25	Prefer not to say	Working Professional	Outside Bangalore	Very High	Yes, I know them well	Not sure	3–6 hours	Studying	Strong passwords	Not sure	Identity theft	I didn't know where to report	Strongly Disagree	Not Sure	No
26–35	Other	Retired	Outside Bangalore	Low	Yes, I know them well	No	More than 6 hours	Online Shopping	Using antivirus	No	Cyberbullying	Not applicable	Strongly Disagree	Not Sure	Maybe
36–50	Male	Retired	North Bangalore	Low	I've heard of them, but not sure	No	Less than 1 hour	Entertainment	Strong passwords	No	Online scam/fraud	No	Strongly Disagree	Not Sure	No
18–25	Male	Retired	Central Bangalore	No Awareness	No, I don't know them	No	More than 6 hours	Banking /UPI	None of the above	No	Hacking	I didn't know where to report	Strongly Disagree	Yes	No

51+	Male	Business Owner	North Bangalore	Low	Yes, I know them well	Not sure	Less than 1 hour	Social Media	Using antivirus	Not sure	None / Prefer not to say	No	Strongly Agree	Yes	Maybe
51+	Female	Other	East Bangalore	Moderate	I've heard of them, but not sure	No	3–6 hours	Work/Business	None of the above	No	Cyberbullying	Not applicable	Strongly Disagree	Yes	Yes
Under 18	Male	Working Professional	East Bangalore	Moderate	I've heard of them, but not sure	Yes	More than 6 hours	Studying	Avoiding unknown links	Yes	OTP/UPI fraud	I didn't know where to report	Strongly Disagree	Yes	Yes
51+	Female	Retired	Outside Bangalore	Moderate	I've heard of them, but not sure	Not sure	More than 6 hours	Social Media	Strong passwords	Not sure	Identity theft	No	Strongly Agree	Not Sure	Yes
18–25	Prefer not to say	Student	Central Bangalore	No Awareness	I've heard of them, but not sure	No	3–6 hours	Entertainment	Avoiding unknown links	No	Online scam/fraud	No	Strongly Disagree	No	Maybe
36–50	Other	Business Owner	Central Bangalore	Low	I've heard of them, but not sure	No	More than 6 hours	Banking /UPI	None of the above	Not sure	Identity theft	Not applicable	Agree	Yes	Not Sure
18–25	Prefer not to say	Homemaker	Central Bangalore	No Awareness	Yes, I know them well	No	3–6 hours	Online Shopping	Using antivirus	Not sure	OTP/UPI fraud	No	Neutral	Not Sure	Maybe
26–35	Female	Homemaker	Central Bangalore	No Awareness	I've heard of them, but not sure	No	1–3 hours	Social Media	Strong passwords	No	Online scam/fraud	Not applicable	Strongly Disagree	No	Not Sure
18–25	Male	Working Professional	Outside Bangalore	Moderate	Yes, I know them well	Not sure	Less than 1 hour	Entertainment	Avoiding unknown links	Not sure	Cyberbullying	I didn't know where to report	Neutral	No	No
26–35	Other	Other	Central Bangalore	Low	No, I don't know them	Yes	3–6 hours	Online Shopping	Two-factor authentication	Not sure	Hacking	Yes	Neutral	Yes	Not Sure
36–50	Other	Other	West Bangalore	Very High	No, I don't know them	Not sure	More than 6 hours	Studying	Two-factor authentication	Not sure	Hacking	Yes	Neutral	No	Yes
26–35	Female	Business Owner	North Bangalore	No Awareness	I've heard of them, but not sure	No	Less than 1 hour	Online Shopping	None of the above	No	Online scam/fraud	I didn't know where to report	Agree	No	Yes
18–25	Prefer not to say	Working Professional	North Bangalore	Low	I've heard of them, but not sure	Yes	More than 6 hours	Banking /UPI	None of the above	Yes	None / Prefer not to say	Not applicable	Strongly Disagree	Yes	Not Sure
51+	Male	Retired	Outside Bangalore	High	Yes, I know them well	No	1–3 hours	Banking /UPI	None of the above	Yes	Online scam/fraud	No	Agree	Not Sure	Not Sure
Under 18	Male	Homemaker	West Bangalore	Moderate	Yes, I know them well	No	1–3 hours	Banking /UPI	Strong passwords	Not sure	Hacking	No	Agree	Not Sure	No

36–50	Female	Homemaker	North Bangalore	Very High	I've heard of them, but not sure	No	Less than 1 hour	Studying	Avoiding unknown links	No	OTP/UPI fraud	I didn't know where to report	Neutral	Yes	Not Sure
51+	Male	Business Owner	East Bangalore	Moderate	I've heard of them, but not sure	No	3–6 hours	Entertainment	None of the above	No	Cyberbullying	Yes	Neutral	No	Maybe
51+	Prefer not to say	Retired	East Bangalore	Very High	Yes, I know them well	Not sure	More than 6 hours	Work/Business	Strong passwords	No	Hacking	No	Neutral	Yes	No
36–50	Male	Business Owner	West Bangalore	Moderate	No, I don't know them	No	Less than 1 hour	Work/Business	Avoiding unknown links	No	Cyberbullying	Not applicable	Strongly Agree	No	No
51+	Prefer not to say	Working Professional	West Bangalore	Low	I've heard of them, but not sure	No	Less than 1 hour	Studying	Strong passwords	Not sure	Hacking	Yes	Neutral	Not Sure	Not Sure
26–35	Male	Retired	Central Bangalore	Moderate	No, I don't know them	Yes	3–6 hours	Entertainment	Strong passwords	No	Identity theft	No	Disagree	No	Yes
26–35	Other	Working Professional	West Bangalore	Low	Yes, I know them well	Not sure	Less than 1 hour	Studying	Strong passwords	Yes	Hacking	I didn't know where to report	Neutral	No	Maybe

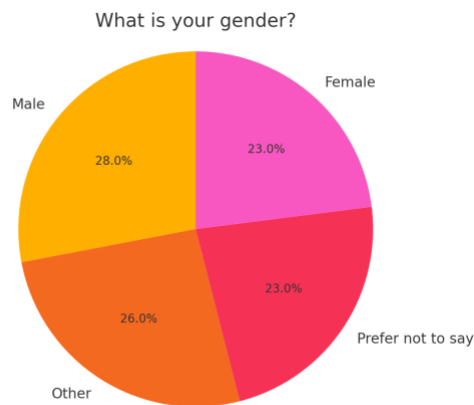
5.2 Statistical Analysis

What is your age group?



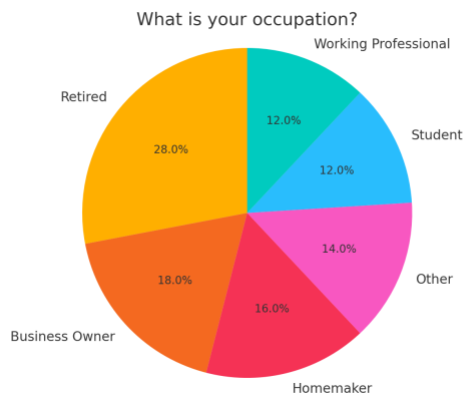
The question "What is your age group?" revealed that the most common response was "18–25" with 26.0%, while the least chosen was "26–35" with 17.0%. Other responses accounted for approximately 57.0% combined. This distribution highlights the general sentiment and awareness levels among the respondents in urban Bangalore.

What is your gender?



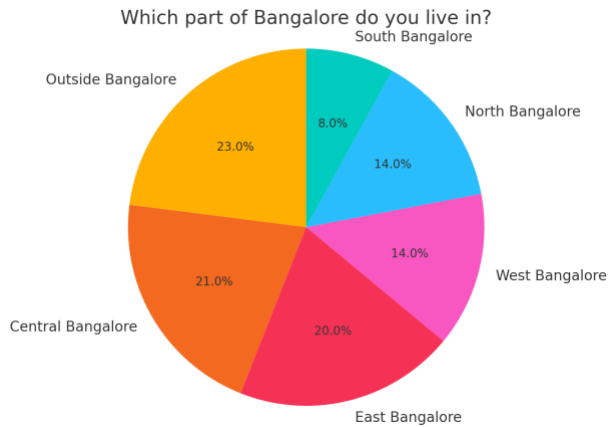
The question "What is your gender?" revealed that the most common response was "Male" with 28.0%, while the least chosen was "Prefer not to say" with 23.0%. Other responses accounted for approximately 49.0% combined. This distribution highlights the general sentiment and awareness levels among the respondents in urban Bangalore.

What is your occupation?



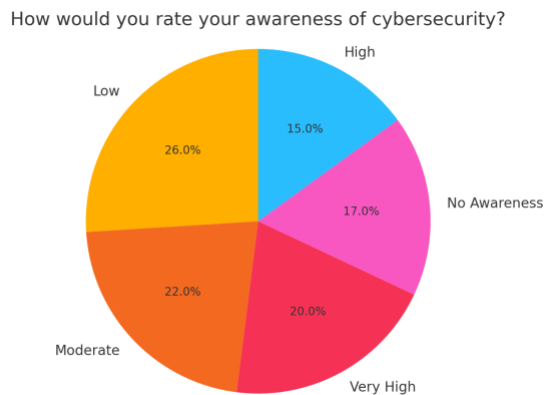
The question "What is your occupation?" revealed that the most common response was "Retired" with 28.0%, while the least chosen was "Student" with 12.0%. Other responses accounted for approximately 60.0% combined. This distribution highlights the general sentiment and awareness levels among the respondents in urban Bangalore.

Which part of Bangalore do you live in?



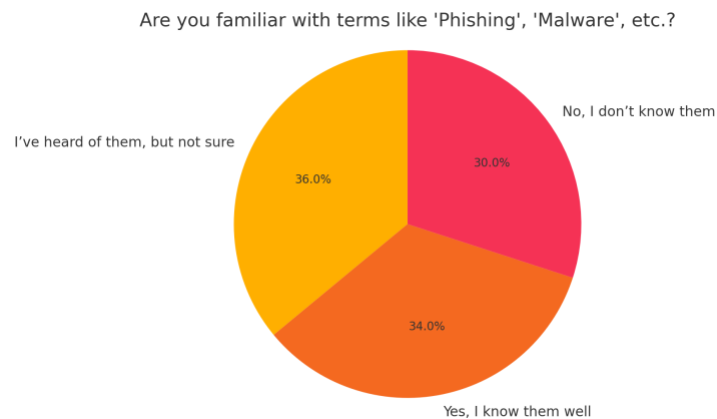
The question "Which part of Bangalore do you live in?" revealed that the most common response was "Outside Bangalore" with 23.0%, while the least chosen was "South Bangalore" with 8.0%. Other responses accounted for approximately 69.0% combined. This distribution highlights the general sentiment and awareness levels among the respondents in urban Bangalore.

How would you rate your awareness of cybersecurity?



The question "How would you rate your awareness of cybersecurity?" revealed that the most common response was "Low" with 26.0%, while the least chosen was "High" with 15.0%. Other responses accounted for approximately 59.0% combined. This distribution highlights the general sentiment and awareness levels among the respondents in urban Bangalore.

Are you familiar with terms like 'Phishing', 'Malware', etc.?



The question "Are you familiar with terms like 'Phishing', 'Malware', etc.?" revealed that the most common response was "I've heard of them, but not sure" with 36.0%, while the least chosen was "No, I don't know them" with 30.0%. Other responses accounted for approximately 34.0% combined. This distribution highlights the general sentiment and awareness levels among the respondents in urban Bangalore.

Have you received formal cybersecurity training?



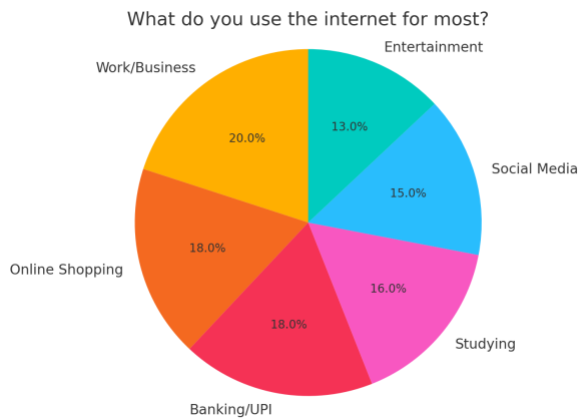
The question "Have you received formal cybersecurity training?" revealed that the most common response was "No" with 42.0%, while the least chosen was "Yes" with 28.0%. Other responses accounted for approximately 30.0% combined. This distribution highlights the general sentiment and awareness levels among the respondents in urban Bangalore.

How many hours do you spend online daily?



The question "How many hours do you spend online daily?" revealed that the most common response was "More than 6 hours" with 34.0%, while the least chosen was "Less than 1 hour" with 20.0%. Other responses accounted for approximately 46.0% combined. This distribution highlights the general sentiment and awareness levels among the respondents in urban Bangalore.

What do you use the internet for most?



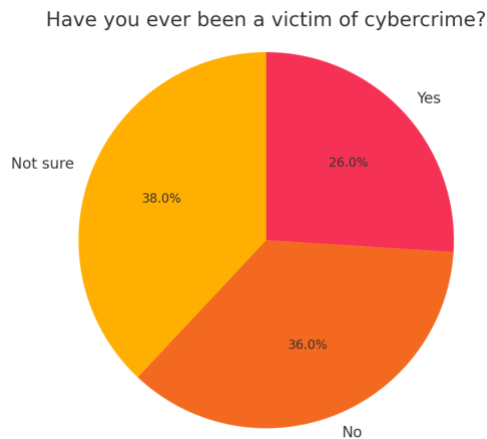
The question "What do you use the internet for most?" revealed that the most common response was "Work/Business" with 20.0%, while the least chosen was "Entertainment" with 13.0%. Other responses accounted for approximately 67.0% combined. This distribution highlights the general sentiment and awareness levels among the respondents in urban Bangalore.

Which cybersecurity practice do you follow most often?



The question "Which cybersecurity practice do you follow most often?" revealed that the most common response was "Avoiding unknown links" with 24.0%, while the least chosen was "Using antivirus" with 16.0%. Other responses accounted for approximately 60.0% combined. This distribution highlights the general sentiment and awareness levels among the respondents in urban Bangalore.

Have you ever been a victim of cybercrime?



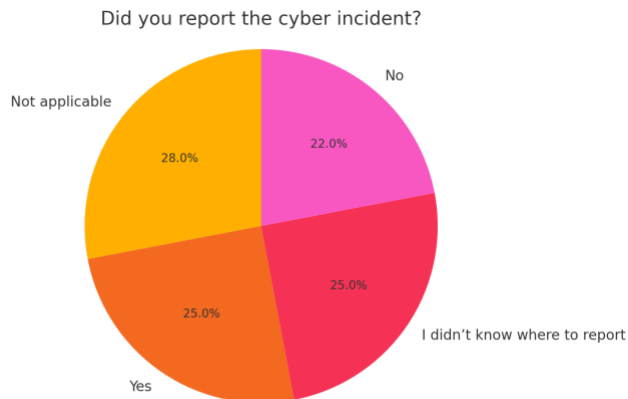
The question "Have you ever been a victim of cybercrime?" revealed that the most common response was "Not sure" with 38.0%, while the least chosen was "Yes" with 26.0%. Other responses accounted for approximately 36.0% combined. This distribution highlights the general sentiment and awareness levels among the respondents in urban Bangalore.

If yes, which of the following best describes it?



The question "If yes, which of the following best describes it?" revealed that the most common response was "Hacking" with 24.0%, while the least chosen was "Cyberbullying" with 14.0%. Other responses accounted for approximately 62.0% combined. This distribution highlights the general sentiment and awareness levels among the respondents in urban Bangalore.

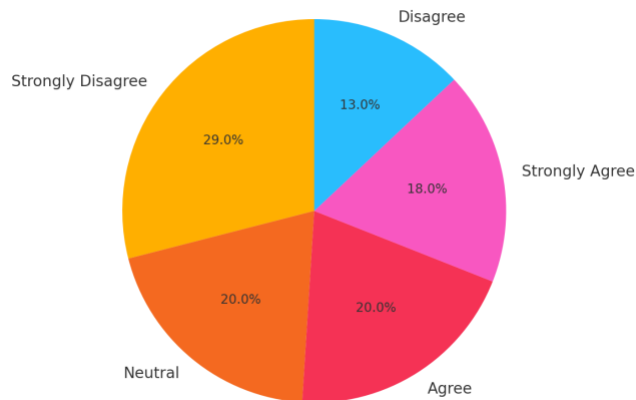
Did you report the cyber incident?



The question "Did you report the cyber incident?" revealed that the most common response was "Not applicable" with 28.0%, while the least chosen was "No" with 22.0%. Other responses accounted for approximately 50.0% combined. This distribution highlights the general sentiment and awareness levels among the respondents in urban Bangalore.

Do you think cybercrimes are increasing in urban Bangalore?

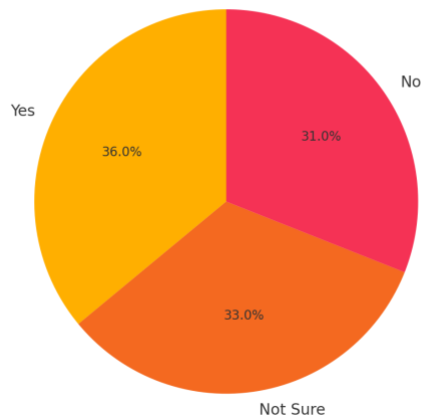
Do you think cybercrimes are increasing in urban Bangalore?



The question "Do you think cybercrimes are increasing in urban Bangalore?" revealed that the most common response was "Strongly Disagree" with 29.0%, while the least chosen was "Disagree" with 13.0%. Other responses accounted for approximately 58.0% combined. This distribution highlights the general sentiment and awareness levels among the respondents in urban Bangalore.

Should the government raise cybersecurity awareness?

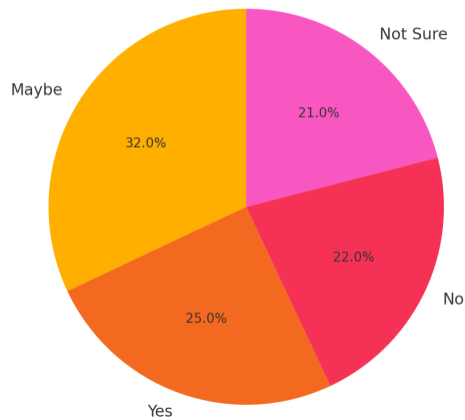
Should the government raise cybersecurity awareness?



The question "Should the government raise cybersecurity awareness?" revealed that the most common response was "Yes" with 36.0%, while the least chosen was "No" with 31.0%. Other responses accounted for approximately 33.0% combined. This distribution highlights the general sentiment and awareness levels among the respondents in urban Bangalore.

Should schools/colleges include cybersecurity education?

Should schools/colleges include cybersecurity education?



The question "Should schools/colleges include cybersecurity education?" revealed that the most common response was "Maybe" with 32.0%, while the least chosen was "Not Sure" with 21.0%. Other responses accounted for approximately 47.0% combined. This distribution highlights the general sentiment and awareness levels among the respondents in urban Bangalore.

6. Challenges and Limitations

6.1. Lack of Public Awareness

Many individuals in urban Bengaluru continue to fall victim to cybercrimes due to inadequate knowledge about online safety practices. Cybercriminals take advantage of common digital behaviors, such as clicking on suspicious links or inadvertently sharing OTPs, to perpetrate fraud. This lack of awareness significantly increases vulnerability, especially among older users and those with limited exposure to cybersecurity education. ^[1]

6.2. Evolving Cybercrime Tactics

Cybercriminals constantly innovate their methods, making it increasingly difficult for law enforcement agencies to stay ahead. The emergence of AI-driven scams, deepfake frauds, and sophisticated phishing attacks signifies a shift toward highly technical and deceptive cyber threats. These tactics are not only harder to detect but also evolve rapidly, outpacing current security protocols and detection tools. ^[1]

6.3. Delayed Law Enforcement Response

The resolution of cybercrime cases often requires a high degree of technical expertise and inter-agency coordination, which can lead to significant delays. Victims frequently report challenges in receiving timely assistance, often due to jurisdictional issues and the complexity of digital evidence gathering. These delays can reduce the chances of recovery and erode public trust in the justice system. ^[1]

6.4. Legal and Regulatory Gaps

India's existing legal framework does not always account for the latest developments in cybercrime, particularly those involving artificial intelligence and cross-border fraud. Cybercriminals often exploit these regulatory blind spots, operating from international locations and using technologies that blur the line between legitimate and illicit digital behavior. This legal ambiguity hampers effective enforcement and prosecution. ^[1]

6.5. Financial and Emotional Impact on Victims

Victims of cybercrimes frequently endure substantial financial losses, many of which are difficult, if not impossible, to recover. Beyond monetary damage, the emotional toll of cyber extortion and online fraud is profound, often resulting in stress, anxiety, and long-term psychological distress. This dual impact underscores the urgent need for both preventive education and victim support services. ^[1]

6.6. Limited Cybersecurity Measures in Small Businesses

Small and medium enterprises (SMEs) in Bengaluru often lack the financial and technical resources to implement comprehensive cybersecurity measures. These businesses are particularly susceptible to data breaches, ransomware attacks, and digital payment frauds. The absence of proper security protocols can lead to severe operational disruptions and long-term reputational damage. ^[1]

7. Conclusion

Cyber crimes in urban Bengaluru are escalating with the rapid digital transformation, posing significant threats to individuals, businesses, and government institutions. Emerging threats such as phishing, ransomware, financial fraud, and identity theft have become more sophisticated, exploiting technological vulnerabilities and human errors.

This study points out key challenges, including the lack of stringent cybersecurity awareness, inadequate law enforcement mechanisms, and evolving attack techniques. While various countermeasures, such as

blockchain-based security, AI-driven threat detection, and robust legal frameworks, have been explored, their effective implementation remains a challenge.

To tackle these growing issues, a multi-stakeholder approach involving government agencies, cybersecurity firms, and public participation is essential. Strengthening legal policies, promoting digital literacy, and investing in advanced security technologies will be crucial in mitigating cyber risks. As urban Bengaluru continues its digital expansion, proactive cybersecurity strategies must be adopted to ensure a safe and resilient cyber ecosystem. ^[1]

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